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PHM–GPT: A Large Language Model for Prognostics and Health Management

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ABSTRACT

In the digital transformation era of the fourth industrial revolution, prognostics and health management (PHM) is playing increasingly important roles in various engineering fields. As the complexity of industrial systems continues to increase, model-based or data-driven PHM technologies face growing challenges related to interpretability, generalization, and applicability, which limit the widespread deployment of PHM technologies. To address these challenges, this PHM–GPT, a large language model (LLM) specifically designed for PHM, is proposed in this paper. By leveraging LLMs, the PHM–GPT unifies anomaly detection, fault diagnosis, and maintenance decision-making tasks, enabling robust generalization across diverse datasets. In detail, a signal–to–text (S2T) pipeline is presented to develop the InsPHM-456k dataset, focusing on representative components such as bearings and gears. Furthermore, a novel framework for adapting general-purpose LLMs into PHM-specific LLMs is proposed through knowledge injection-based pretraining, PHM-specific instruction tuning, and downstream application fine-tuning during PHM. Additionally, an efficient architecture is introduced to incorporate low-rank adaptation adapters into the group attention module and the feedforward neural network. To validate the effectiveness of the PHM–GPT, extensive simulation studies are conducted, demonstrating its strong generalization across diverse datasets and its broad applicability to machinery components such as bearings and gears. Beyond automatically providing anomalies, diagnoses, and maintenance results, the PHM–GPT exhibits emergent abilities that have never been observed before, such as attribution reasoning, threshold setting, and knowledge discovery, which contribute to enhanced qualitative interpretability and deeper insight into system behaviors. Finally, this paper provides new insights into the PHM field and explores the future of LLMs in terms of advancing PHM technology deployment, accelerating the digital transformation process during the fourth industrial revolution.

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1. Introduction

Prognostics and health management (PHM) has become a critical element in intelligent manufacturing, including cases involving space station, aero engines, high-speed trains, and wind turbines [1–5]. The advancement of PHM enhances the ability to detect potential equipment anomalies, facilitates the ability to diagnose complex faults, improves the accuracy of remaining useful life (RUL) predictions, and shifts regular maintenance toward predictive maintenance decisions. In the future, PHM may fully leverage both real-time and historical data to enable comprehensive maintenance management, thereby ensuring the safe operation of equipment, reducing product lifecycle costs and increasing the availability of complex systems [6–8].

As early as the 1990s, machine learning methods, such as expert systems [9], support vector machines [10], random forests [11], and *k*-nearest neighbors [12], were extensively applied to PHM tasks. These methods began to establish relationships between the extracted features and the health state of structures, systems, or machines, reducing the reliance on human labor during PHM [13]. However, as data increase in complexity and scale, the limitations of machine learning methods become apparent, making it

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challenging to automatically process large-scale data and leading to heavy reliance on manually extracted features for attaining optimal performance. Deep learning (DL), as a mainstream branch of machine learning, has brought new momentum to the field of PHM since the 2010s [14]. Popular DL architectures, such as convolutional neural networks (CNNs) [15], recurrent neural networks (RNNs) [16], and transformer models [17,18], have achieved powerful performance in various PHM tasks, including anomaly detection, fault diagnosis, and RUL prediction. These methods aim to provide end-to-end solutions for specific PHM tasks in response to the growing volume of monitoring data. Compared with the traditional machine learning methods, DL methods not only automatically extract features from the input raw data, but also improve the performance attained in various PHM tasks [19,20], thereby increasing the intelligence of manufacturing.

As DL-based methods continue to develop, new challenges are gradually emerging [6,21]. First, interpretability remains a fundamental limitation of DL models in PHM [22]. Although DL models have strong representation capabilities, their "black-box" nature makes the PHM-related decision-making process difficult to explain, which hinders users' trust in the results. While some studies [23,24] have successfully improved the interpretability of DL models in PHM, this rare interpretability often struggles to keep up with the rapid development of more advanced DL architectures and remains incompatible with other methods such as transfer learning and continual learning. Second, generalization remains a challenge when applying DL models in PHM [25–27]. Transfer learning methods such as data augmentation, domain adaptation, and adversarial training have been proposed to address domain shifts [28]. However, these methods require the careful selection of source–target pairs and often need labeled data in the target domain. This dependency greatly limits their effectiveness when addressing previously unseen or unlabeled data, which is a common scenario in real-world applications. Finally, the applicability issue hinders the deployment of models in real-world industrial environments [29]. While the current models perform remarkably well in laboratory settings, their results are still unsatisfactory when deployed in practical settings. For instance, a diagnostic model trained for bearings often fails to perform reliably when it is applied to other components such as gears. Moreover, DL-based PHM methods are typically developed for specific tasks, often requiring task-specific architectures and independent training procedures. As a result, they tend to lack adaptability when applied to diverse PHM tasks. In summary, because of their inherent limitations, the potential of DL-based PHM methods has not yet been fully realized.

Currently, large language models (LLMs) such as generative pretrained transformer 4 (GPT-4) [30], and large language model meta artificial intelligence 3 (LLaMA-3) [31] have demonstrated remarkable capabilities, approaching or even exceeding human-level capabilities in an increasing number of fields. In natural language processing tasks, LLMs exhibit powerful text generation and comprehension abilities, with applications in machine translation [32], text summarization [33], and sentiment analysis [34]. In scientific research, LLMs help discover new theories in mathematics, physics, and chemistry, and assist in proving theorems [35]. In health care, LLMs provide enhanced clinical efficiency by supporting tasks such as medical record analysis, clinical decision-making, and drug discovery [36]. In soft engineering cases, LLMs facilitate software development, autonomously generate code and assist with lifecycle energy [37,38]. Building on the remarkable progress achieved by LLMs, their integration into PHM systems offers an exciting opportunity to address the challenges that are faced by DL-based PHM methods.

While DL-based PHM methods provide high accuracy on specific datasets and independent tasks, they often struggle to generalize across different equipment, working conditions, and PHM tasks. In contrast, LLMs evolve the paradigm of PHM by introducing a unified model that is capable of simultaneously addressing multiple PHM tasks, as shown in Fig. 1. By benefitting from their large-scale pretraining operations conducted on massive and diverse datasets, LLMs exhibit strong generalization capabilities. Furthermore, their strengths in representation, understanding, reasoning, and knowledge discovery enable them to significantly enhance the performance attained in complex industrial settings. For example, LLMs can combine external input knowledge with internal reasoning capabilities, resulting in more interpretable predictions, which are often lacking in black-box DL models. Furthermore, when DL-based models trained on vibration signals fail to identify unknown or compound faults because of their limited scopes, LLMs can infer the most likely fault by referencing similar historical cases, thereby enhancing their adaptability and practical usability in real-world applications. In summary, this paradigm evolution not only addresses the limitations of DL-based methods but also unlocks new possibilities for more scalable and efficient PHM techniques in the era of intelligent manufacturing.

The aim of this paper is to explore how LLMs can drive the evolution of PHM from data-specific DL models to a unified large model, addressing challenges such as interpretability, generalization, and applicability. To the best of our knowledge, the PHM–GPT is the first LLM-based framework to be applied to PHM in a unified, multitask learning setting, encompassing diverse components such as bearings and gears, and it is evaluated across a wide range of datasets. The main contributions of this study are summarized as follows:

A signal-to-text (S2T) pipeline that incorporates prior knowledge of signal processing knowledge to transform time series signals into text-based inputs that are suitable for LLMs is proposed. Building upon this pipeline, the InstPHM-456k dataset is constructed; it comprises instruction tuning subsets for detection,

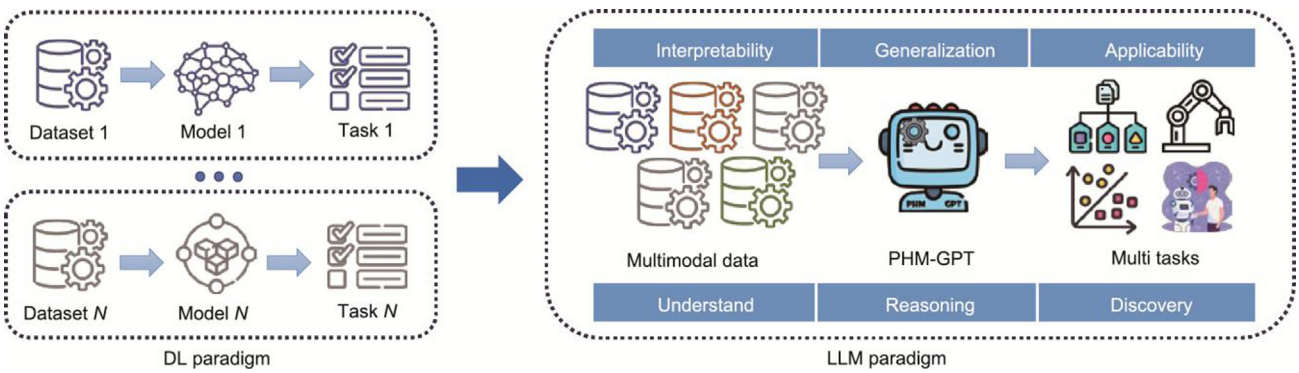


Fig. 1. The paradigm evolution of PHM from deep learning to large language models. *N*: number.

diagnosis, and maintenance tasks, as well as a multiround dialog subset that supports interaction across the entire PHM system.

A novel framework is proposed for adapting general-purpose LLMs to PHM-specific LLMs through a three-stage process: knowledge injection-based pretraining, PHM-specific instruction tuning, and downstream application fine-tuning. In addition, an efficient model architecture is introduced by integrating low-rank adaptation (LoRA) adapters into both a group attention module and a feedforward neural network.

An extensive evaluation of the PHM–GPT across multiple simulation settings highlights its strong performance. By leveraging the powerful capabilities of LLMs, the PHM–GPT can not only automatically obtain detection, diagnosis, and maintenance results, but also exhibit emergent abilities that have never appeared in traditional PHM systems, such as attribution reasoning, threshold setting, and knowledge discovery.

The paper is structured as follows. Section 1 introduces the research background and objectives. Section 2 reviews the related works in the field of LLMs and PHM. Section 3 describes the PHM instruction dataset and its construction method. Section 4 presents a novel framework for the fine-tuning LLMs in PHM in a supervised manner. Section 5 describes the simulation setup and results, emphasizing the performance achieved across various tasks. Section 6 discusses the emergent abilities of the PHM–GPT, along with its limitations. Finally, Section 7 concludes the paper and outlines directions for future work.

2. Related works

In this section, a comprehensive literature review is presented from three perspectives. First, a review of time series models that are widely applied in PHM is provided, along with a discussion of their limitations. Second, the development trends of LLMs are explored, with an emphasis on the distinction between general-purpose and domain-specific LLMs. Finally, the applications of LLMs in PHM are summarized. Collectively, these perspectives provide the technical background for our work.

2.1. Application of time-series models in PHM

In recent years, DL-based time series models [21] have been widely used in PHM because of their ability to automatically learn complex patterns from monitoring data. Architectures such as CNNs, RNNs, and transformers have demonstrated strong performance. However, the ideal assumption of being independent and identically distributed is often not satisfied in industrial practice, leading to poor generalization [25,26]. For example, models trained on bearing data in laboratory settings often fail in real-world applications because of the presence of unseen fault modes and different signal characteristics. While domain adaptation has been proposed to alleviate such issues, it typically relies on the availability of labeled target data and well-aligned feature distributions. Alternatively, physics-informed neural networks aim to provide improved generalization by embedding physical principles into the model design and learning procedures. However, they often require extensive domain knowledge, and their effectiveness diminishes when system behavior deviates from the underlying physical assumptions. All these limitations underscore the need for more flexible and scalable solutions.

Recent advances in large-scale time-series models, such as RmGPT [39] and BearingFM [40], have attracted attention in PHM, with the primary focusing being on fault diagnosis tasks. These models utilize extensive and diverse vibration signal datasets for pretraining on a transformer architecture to achieve performance breakthroughs. However, one key concern is that the scales and diversity levels of the datasets used to train LLMs far surpass those used to train large time series models. This raises questions about whether the scaling laws observed in LLMs can be effectively applied to limited industrial datasets and smaller model architectures. Another concern is that time-series models are not well suited for addressing maintenance-related tasks because their decision spaces are limited. These tasks include spare parts management, and task planning. The final concern is whether the demand for interpretability in industrial scenarios can be addressed through foundational time series models. There is insufficient evidence to determine whether interpretability and generalization can be compatible in such models.

2.2. General-purpose and domain-specific LLMs

LLMs, including GPTs, LLaMA, and their derivatives, have achieved remarkable success in natural language processing tasks, showcasing advanced capabilities such as multitask learning, contextual reasoning, and knowledge extraction. These LLMs can be broadly categorized into two primary areas: general-purpose and domain specific LLMs.

General-purpose LLMs refer to the pretrained models that are built on vast amounts of data from diverse domains, enabling them to handle a wide range of tasks. Their emergent abilities, such as chain-of-thought reasoning and few-shot learning, enable these models to adapt to diverse downstream tasks with minimal retraining. Examples include GPT-4 [30], LLaMA-3 [31], and Qwen-2.5 [41]. These models are not limited to a specific domain and can perform tasks such as summarization, question answering, and translation, computation across diverse domains. With the rapid development of their multimodal processing capabilities, general-purpose LLMs are evolving from single language models to intelligent systems that support complex scenarios and multitask collaboration, providing a practical pathway for achieving artificial general intelligence. However, their hallucinations and other reliability issues may limit their performance in specialized domains.

In response to these issues, domain-specific LLMs refer to fine-tuned models that are built on specialized datasets focusing on domain-relevant knowledge to address the unique challenges of each domain. For example, models such as StructGPT [42] for text summarization, CodeGen [43] for code generation, BioGPT [44] for biomedical applications, and ChatLaw [45] for legal text analysis illustrate their diverse capabilities. These models leverage the core capabilities of general-purpose LLMs, improving the accuracy achieved when addressing domain-specific needs and thereby highlighting the importance of developing domain-specific LLMs that are tailored for PHM-related tasks.

2.3. Application of LLMs in PHM

Despite the transformative potential of LLMs in various domains, their applications in PHM are in their early stages. PHM tasks such as anomaly detection, fault diagnosis, and maintenance decision-making require the handling of diverse data types (e.g., vibration signals and text logs), presenting unique challenges for large models.

A variety of models have been proposed to apply LLMs to fault diagnosis work. For example, FCLLM-DT [46] integrates the digital twin methodology and retrieval augmented generation (RAG)-assisted LLMs to improve the accuracy of bearing fault diagnoses. FD-LLM [47] and LLM-BF [48] were designed for fault diagnosis tasks by formulating the training process of LLMs as multiclass classification problems. These models adapt LLMs to handle numerical data inputs, enabling the identification of various faults from time series sensor data. PaInfoGPT [49] is an LLM-based

diagnosis framework that operates under partial information, leveraging a mutual information-based masking strategy for pretraining and employing a parallel side-adapter for fine-tuning. All the aforementioned models establish classification output heads but do not account for the interpretability of their results when implementing generalization.

For this reason, FaultExplain [50] was specifically designed as an interactive tool to improve fault detection results, diagnoses, and explanations. In addition to providing interpretable results, VSLLaVA [51] was established as a multimodal diagnosis pipeline that uses signal–question–answer triplets to fine-tune contrastive language image pretraining (CLIP) and LLM. However, while these models offer explainable results, their generalizability has not yet been fully addressed, and they have not been applied to other PHM tasks. LLM-R [52] was proposed as a maintenance scheme generation method based on LLMs, with the aim of enhancing the intelligence of maintenance planning and introducing innovative technical approaches for equipment maintenance. BearLLM [53] focuses on the comprehensiveness of PHM by unifying multiple bearing-related tasks, such as detection, diagnosis, and maintenance, through the processing of user prompts and vibration signals. However, each PHM task remains relatively independent, and a common multiround dialog mechanism that is typically seen in LLMs has not yet been developed. In addition, some of the inputs of LLMs are intermediate representations derived from trained DL models, which are inherently more challenging to interpret.

2.4. Research gaps and motivations

The research interests and characteristics based on the above literature are summarized in Table 1 [44–53]. The existing time series models were typically designed for specific tasks such as fault diagnosis. These models lack flexibility and are not easily extendable to other PHM tasks, such as detection or maintenance. Although language-based models have the potential to handle multiple PHM tasks, most existing models still focus primarily on diagnosis. Despite the growing demand for integrated diagnostic and decision support tools, most of these models have not been designed to handle multiple PHM tasks in a unified framework. Other critical gaps lie in interpretability, generalization, and applicability. Although some models exhibit promising generalization performance, the small sizes of datasets often lead to overfitting and poor generalization to unseen or incomplete data. Moreover, most existing models have been developed using bearing fault data, which significantly limits their applicability. For instance, a model trained on bearing data under laboratory settings may fail to diagnose faults in a gearbox under industrial settings. Furthermore, their interpretability remains insufficient, as many models prioritize the accuracy of their results without providing clear explanations, which restricts their trustworthiness in safety–critical industrial applications.

To fill the above research gaps, this study is aimed at developing a comprehensive framework that integrates various PHM tasks, such as detection, diagnosis, and maintenance, in a real-time interactive manner. It dynamically adapts to new working conditions and incomplete input data, and ensures interpretability. Moreover, the impressive emergent abilities of LLMs, such as understanding, reasoning, and knowledge discovery, have not been fully explored in many current PHM models.

This study seeks to leverage these emergent abilities to further enhance the generalizability, interpretability, and adaptability of the developed model, ultimately improving the trustworthiness and effectiveness of PHM systems in safety–critical applications, particularly in the era of intelligent manufacturing.

3. InstPHM-456k dataset and its construction process

This section presents the S2T pipeline and the methodology used to construct the InstPHM-456k instruction dataset. These steps are essential for facilitating effective multitask learning with the PHM–GPT, covering various tasks such as anomaly detection, fault diagnosis, and maintenance decision-making.

3.1. S2T pipeline

In the traditional DL-based PHM models, raw sensor data are processed in an end-to-end manner, especially in detection and diagnosis tasks, where the models analyze time series data for prediction. However, with the rise of LLMs, raw sensor signals need to be converted into text descriptions. In practice, signals are collected from various devices, sensors, and analog-to-digital converters, resulting in varying sampling rates, amplitudes, and time window lengths.

To address the complexity of raw sensor signals and achieve a unified signal-to-text representation, prior signal processing knowledge is incorporated into the feature extraction process. The basic features are extracted directly from the raw signals to capture their fundamental statistical characteristics, without relying on complex signal processing techniques. As shown in Table S1 in Appendix, a total of 24 basic features are computed, consisting of 12 time-domain features and 12 frequency-domain features, including the mean, standard deviation, root mean square error, skewness, kurtosis, crest factor, clearance factor, shape factor, and impulse factor. Each of these features serves a specific purpose for PHM. For example, the physical meanings of the crest factor and impulse factor are sensitive to the detection of impacts in a signal, indicating the presence of sudden disturbances or shocks. The clearance factor is typically used to assess equipment wear. Additionally, kurtosis reflects the impact characteristics of vibration signals, highlighting the presence of sharp peaks or abnormal events. The key advantage of this representation method is its

Table 1
Related studies on large models for PHM.

Model	Description	Research focus			Interpretability	Generalization	Applicability	
		Detection	Diagnosis	Maintenance			Bearing	Gear
RmGPT [44]	Time series	–	*P*	–	–	*P*	*P*	*P*
BearingFM [45]	Time series	–	*P*	–	–	*P*	*P*	–
FCLLM-DT [46]	Language	–	*P*	–	–	–	*P*	–
FD-LLM [47]	Language	–	*P*	–	–	*P*	*P*	–
LLM-BF [48]	Language	–	*P*	–	–	*P*	*P*	–
PaInfoGPT [49]	Language	–	*P*	–	–	*P*	*P*	–
FaultExplain [50]	Language	–	*P*	–	*P*	–	–	–
VSLLaVA [51]	Language	–	*P*	–	*P*	–	–	–
LLM-R [52]	Language	–	–	*P*	–	–	–	–
BearLLM [53]	Language	*P*	*P*	*P*	–	*P*	*P*	–
PHM–GPT	Language	*P*	*P*	*P*	*P*	*P*	*P*	*P*

ability to effectively capture the fundamental characteristics of potential mechanical anomalies and faults using only a few feature values.

Moreover, owing to the distinct mechanical principles and fault mechanisms of different components, some unique features have been developed for bearings and gears. For bearings, fault characteristic frequencies are produced by periodic impulses, which vary depending on the locations of faults [54]. These frequencies, such as the ballpass frequency outer race (BPFO), ballpass frequency inner race (BPFI), ball spin frequency (BSF), and fundamental train frequency (FTF), are calculated via the formulas in Table 2. In our pipeline, the amplitude values produced at these specific frequencies are extracted as bearing-specific features, as they effectively indicate the presence and types of localized faults. Notably, these values are obtained by the envelope signal after applying filtering.

In contrast, gears exhibit different fault characteristics because of their meshing behavior and rotational dynamics. Normal signals tend to be dominated by the mesh frequency and its harmonics, and are superimposed on a noise floor that is generally assumed to have a Gaussian distribution. When a gear fault occurs, it usually leads to significant changes in the sidebands located near the mesh frequency. To capture these changes, a range of gear-specific features [55] have been developed, as shown in Table 3. FM0 can detect the major changes exhibited by the gear meshing pattern. FM4 is sensitive to the detection of faults that are isolated to a few teeth, which is defined as the kurtosis of the difference signal. M6A is designed to detect surface damage but gives more weight to high-power peaks. NA4 is computed by dividing the fourth statistical moment of the residual signal by the square of its run time averaged variance. In the future, the S2T pipeline can benefit from additional features derived from advanced signal processing methods such as wavelet packet decomposition, enhancing the capabilities of large models.

Table 2
Bearing-specific features in the S2T pipeline.

Frequency	Bearing-specific feature
BPFO	$\frac{ef_r}{2}\left(1-\frac{d}{D}\cos\phi\right)$
BPFI	$\frac{ef_r}{2}\left(1+\frac{d}{D}\cos\phi\right)$
FTF	$\frac{f_r}{2}\left(1-\frac{d}{D}\cos\phi\right)$
BSF	$\frac{D}{2d}\left\{1-\left(\frac{d}{D}\cos\phi\right)^2\right\}$

e is the number of rolling elements, f_r is the shaft speed, d is the diameter of the rolling element, D is the bearing diameter, and ϕ is the angle of the load from the radial plane.

Table 3
Gear-specific features in the S2T pipeline.

Gear-specific feature	Gear-specific feature
FM0	$\frac{\mathrm{PP}(x(n))}{\sum_{h=0}^{H}P_h}$
FM4	$\frac{N\sum_{n=1}^{N}\left(d(n)-\bar{d}\right)^4}{\left[\sum_{n=1}^{N}\left(d(n)-\bar{d}\right)^2\right]^2}$
M6A	$\frac{N^2\sum_{n=1}^{N}\left(d(n)-\bar{d}\right)^6}{\left[\sum_{n=1}^{N}\left(d(n)-\bar{d}\right)^2\right]^3}$
NA4	$\frac{N\sum_{n=1}^{N}\left(r_{nM}-\bar{r}_M\right)^4}{\left\{\frac{1}{M}\sum_{m=1}^{M}\left[\sum_{n=1}^{N}\left(r_{nm}-\bar{r}_m\right)^2\right]\right\}^2}$

$\mathrm{PP}(x(n))$ is the maximum peak-to-peak amplitude of the signal $x(n)$, P_h is the amplitude of the hth harmonic, H is the total number of harmonics in the frequency range, $\bar{d}$ is the mean of the difference signal $d(n)$, N is the number of points in the time signal, $\bar{r}$ is the mean of the residual signal, M is the number of the current time signal, m is the index of the time signal in the run ensemble, r_{nm} or r_{nM} is the residual signal value at the nth data point of the mth or Mth signal in the ensemble, respectively, and $\bar{r}_m$ is its mean value.

3.2. Instruction dataset construction method

The process of constructing the instruction dataset for the PHM relies on multiple data sources. A substantial amount of high-quality monitoring data is collected from sensors, including vibrations, and sounds, with high resolutions and low noise levels. Additionally, maintenance text data include service manuals, maintenance logs, and operational records, detailing the diagnostic procedures and solutions employed for equipment failures. However, the volume of these data is relatively limited. A feasible strategy is to generate maintenance data using GPTs or other LLMs, which simulate device failure scenarios and maintenance procedures.

Raw data collected from monitoring signals and maintenance text sources cannot be directly utilized and require a series of pre-processing steps. For monitoring signals, preprocessing begins with cleaning, which involves removing invalid data such as missing values and outliers. Next is splitting, where a sliding window is employed to partition signals into segments. The selection of the window size and stride depends on the specific data and is designed to capture the patterns that most accurately reflect the status of the target device. Finally, the S2T pipeline is applied to transform the segments into meaningful feature representations, enabling their seamless integration with LLMs. In contrast, textual data require less extensive preprocessing work because they are already in language format. The primary focus is on cleaning, including the removal of stop words, punctuation, and irrelevant content to retain only meaningful information. Notably, the maintenance data generated by LLMs require verification by experts to confirm their realism and validity.

The instruction template plays a central role in guiding the LLM to understand task objectives and generate accurate responses. For a specific PHM task, the instruction template can be reused across different datasets, which reduces the need for extensive prompt engineering. Each template designed for a single PHM task consists of three parts: system prompts, an instruction input, and a model output. The instruction input integrates task-specific instructions, domain-related descriptions and statistical features obtained from the S2T pipeline.

As illustrated in Fig. 2, the template adopts a unified format to guide the model in completing the corresponding tasks. For example, in the detection task, the template directs the LLM to determine whether the given component is abnormal. In diagnosis tasks, the template focuses on identifying specific fault. Notably, for both detection and diagnosis tasks, the responses of LLMs are constrained within a limited scope. For the detection task, the instruction only requires an answer to the question of whether

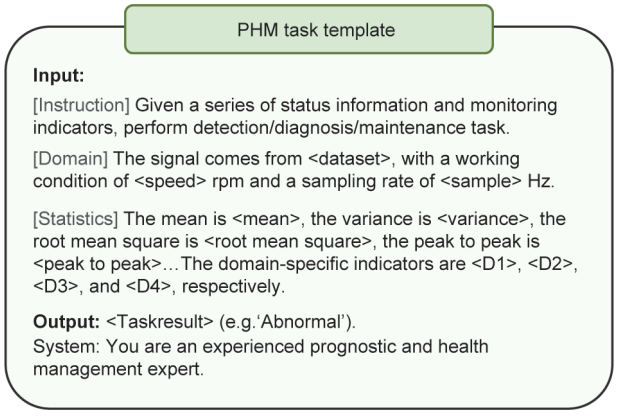


Fig. 2. The instruction template for PHM task. rpm: revolutions per minute.

an anomaly exists, whereas for the diagnosis task, the instruction only asks the LLM to identify the fault type. By limiting responses to simple outputs, the template helps reduce the hallucinations produced by LLMs and improves their performance. In the maintenance task, the template replaces the domain-related descriptions and statistical features with diagnostic results as input, guiding the LLM to generate maintenance recommendations. To align with practical PHM scenarios, a multiround dialog template has also been developed, as illustrated in Fig. 3. Apart from the system prompt, instruction input, and model output parts, this template incorporates a dialog history, enabling the LLM to maintain contextual understanding across multiple interaction rounds.

3.3. InstPHM-456k dataset construction

Despite the availability of numerous public datasets, most of them are limited to single-task evaluations within individual datasets and fail to account for broader tasks, different components, various working conditions, different sampling rates, and sufficient data volume. To address these issues, a large-scale dataset is constructed by integrating data from 15 publicly available sources: Case Western Reserve University (CWRU) bearing dataset, Dynamics and Identification Research Group (DIRG) bearing dataset, Harbin Institute of Technology (HIT) bearing dataset, Huazhong University of Science and Technology bearing (HUB) dataset, Intelligent Maintenance Systems (IMS) bearing dataset, Jiangnan University (JNU) bearing dataset, Paderborn University (PU) bearing dataset, Shandong University of Science and Technology bearing (SDUB) dataset, Southeast University bearing (SEUB) dataset, Xi'an Jiaotong University and Sumyoung Technology (XJTU) bearing dataset, University of Connecticut (UOC) gear dataset, Southeast University gear (SEUG) dataset, Shandong University of Science and Technology gear (SDUG) dataset, Huazhong University of Science and Technology gear (HUG) dataset, and Beijing Jiaotong University (BJTU) gear dataset. This integration scheme forms the foundation of the InstPHM-456k dataset, offering a diverse and extensive collection of data. The distribution of the data sources is illustrated in Fig. 4, where the outer circle displays the number of instances contributed by each dataset, while the inner ring, segmented by color intensity, indicates the proportion of each health state. In total, 114 236 vibration signal instances are included. Each instance is paired with four instruction samples corresponding to detection, diagnosis, maintenance, and PHM dialog tasks, resulting in a total of 456 944 instructions—thus giving the dataset its name: InstPHM-456k.

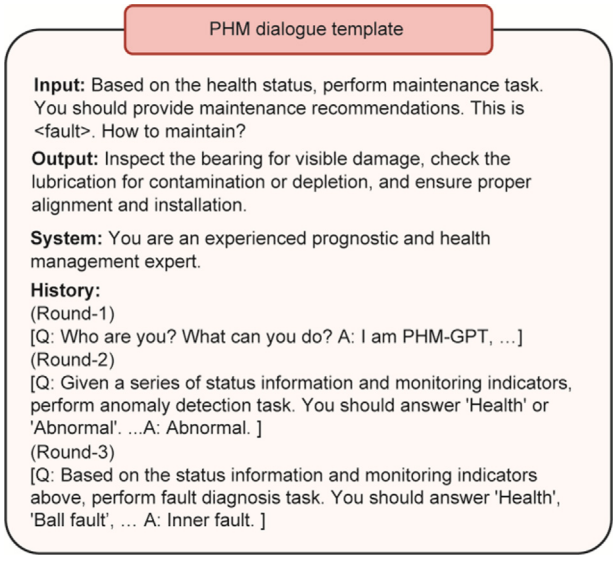


Fig. 3. The instruction template for PHM dialog.

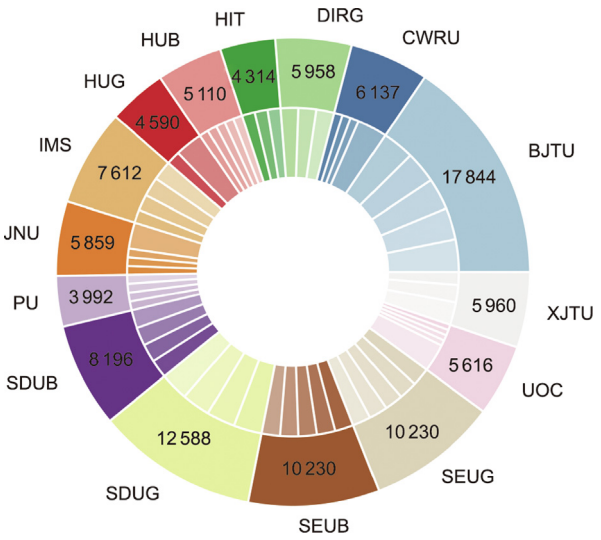


Fig. 4. The dataset source distribution of InstPHM-456k.

For each instance, after conducting preprocessing, statistical features are extracted using the S2T pipeline, which converts time-series signals into textual representations. These representations are then embedded as part of the instruction input, allowing the LLMs to process them and respond. For single PHM tasks such as detection and diagnosis, the signal features and corresponding task-specific instructions are input according to the predefined template. For the multiround PHM dialogue task, the template is extended with a dialogue history. Importantly, both single- and multi-round tasks share the same model architecture, enabling unified learning across diverse PHM tasks. The final instruction-tuning dataset is constructed by integrating data derived from multiple sources through a standardized process that includes preprocessing, transformation via the S2T pipeline, template-based generation, and formatting, as illustrated in Fig. 5.

4. Framework for generalizing the PHM–GPT from pretraining to downstream tasks

4.1. Model architecture design

The current mainstream LLMs, such as GPTs and LLaMA, are based on a decoder-only architecture, which removes the need for an encoder and involves relying solely on the decoder to generate the target sequence. By employing causal language modeling through autoregressive regression, this architecture focuses on predicting the conditional probability distribution of the current token. Its causal masking mechanism ensures that each token involved in the generation task depends only on its previous token, making the model more adaptable and easier to transfer across diverse downstream tasks.

On the basis of the decoder-only architecture employed by the LLMs, an efficient fine-tuning model architecture that incorporates low-rank adaptation (LoRA) to enhance the performance achieved in causal language modeling tasks while significantly reducing both the data and computational requirements is designed. For any deep neural network, the model parameters θ can be represented as the sum of their initial values θ_0 and the subsequent gradient descent updates are learned on minibatches of the training data. The gradient descent updates from epoch $t-1$ to epoch t can be expressed as

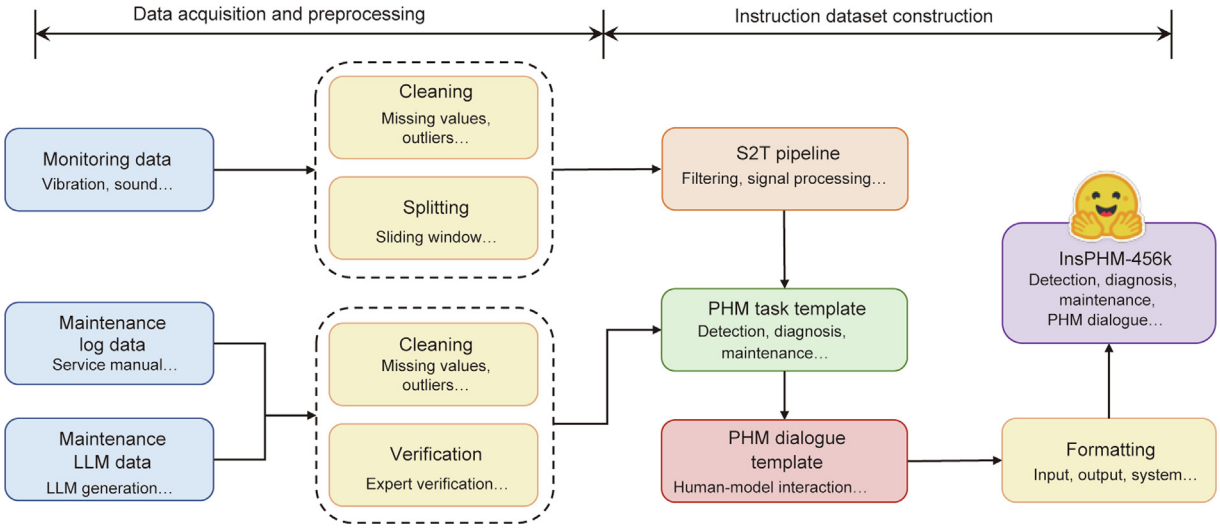


Fig. 5. The instruction dataset construction method of InstPHM-456k.

$$\theta_t \leftarrow \theta_{t-1} - \alpha \nabla_\theta \mathscr{L}|_{\theta=\theta_{t-1}} \tag{1}$$

where α is the learning rate, ∇ is the computed gradient, and $\mathscr{L}$ is the loss function. After T epochs, the trained model parameters θ_T can be expressed as follows:

$$\theta_T = \theta_0 - \alpha \nabla_\theta \mathscr{L}|_{\theta=\theta_0} - \cdots - \alpha \nabla_\theta \mathscr{L}|_{\theta=\theta_{T-1}} \tag{2}$$

From this perspective, a fine-tuned model can be understood as a set of model parameters that are further refined by aggregating additional gradient updates on a specialized dataset as follows:

$$\theta_F = \theta_0 - \alpha \sum_{t=0}^{T-1} \nabla_\theta \mathscr{L}|_{\theta=\theta_t} - \alpha \sum_{t=T}^{T+F} \nabla_\theta \mathscr{L}|_{\theta=\theta_t} \tag{3}$$

where F indicates the number of fine-tuning epochs and θ_F represents the fine-tuned model parameters. Thus, the pretraining and fine-tuning processes can be decomposed into two distinct components, with the decomposition achieved through two separate sets of model parameters. This is the core idea behind LoRA: introducing new weight parameters specifically for fine-tuning, while keeping the original pretraining parameters unchanged. To further reduce the computational cost and memory usage incurred by the large model, these new parameters can be represented through a low-rank matrix approximation, which is achieved by two new matrices, $\boldsymbol{A}$ and $\boldsymbol{B}$.

Finally, the decoder block incorporating the LoRA adapter consists of two modules, such as, a group attention (GA) module and a feedforward neural network (FFN), as illustrated in Fig. 6. In the GA module, the query, key, and value weight are central to capturing the global dependencies of the given sequence. For fine-tuning, low-rank matrices $\boldsymbol{A}_{\mathrm{ATT}}$ and $\boldsymbol{B}_{\mathrm{ATT}}$ are introduced into the pretrained weights W_q, W_k, and W_v to approximate updates. Additionally, rotary position embedding is introduced to capture the relative positional information contained in long sequences, and a key–value (KV) cache is incorporated to avoid redundant computations. In the FFN module, the pretrained weights $\mathrm{W}_{\mathrm{FFN}}$ remain frozen, while two low-rank learnable matrices $\boldsymbol{A}_{\mathrm{FFN}}$ and $\boldsymbol{B}_{\mathrm{FFN}}$ are introduced to restrict the weight updates to a low-rank subspace. Both modules utilize the root mean square normalization (RMS norm) to ensure a stable gradient flow, while residual connections preserve the flow of input information.

4.2. Three-stage training strategy for the PHM–GPT

LLMs have achieved remarkable success in various domains, demonstrating powerful capabilities in terms of both understanding and generation. However, their applications in PHM face significant challenges because of the lack of domain-specific knowledge and the limited adaptability of these models to specialized PHM tasks. In this section, a systematic framework is proposed for adapting general-purpose LLMs to PHM-specific LLMs, as shown in Fig. 7. The overall framework follows a three-stage training strategy: knowledge injection-based pretraining, PHM-specific instruction tuning, and downstream application fine-tuning for PHM.

In the pretraining stage, the existing LLMs (e.g., GPTs, LLaMA, and Claude) are employed as backbone models to create domain LLMs by injecting PHM knowledge. This PHM knowledge is sourced from a variety of materials, including academic papers, specialized books, and equipment maintenance manuals. This high-quality text knowledge is systematically processed and integrated into the semantic space of the large model, enabling it to understand and generate content that is specific to the PHM domain.

Following the pretraining stage, an instruction tuning stage is performed to improve the performance of the model on

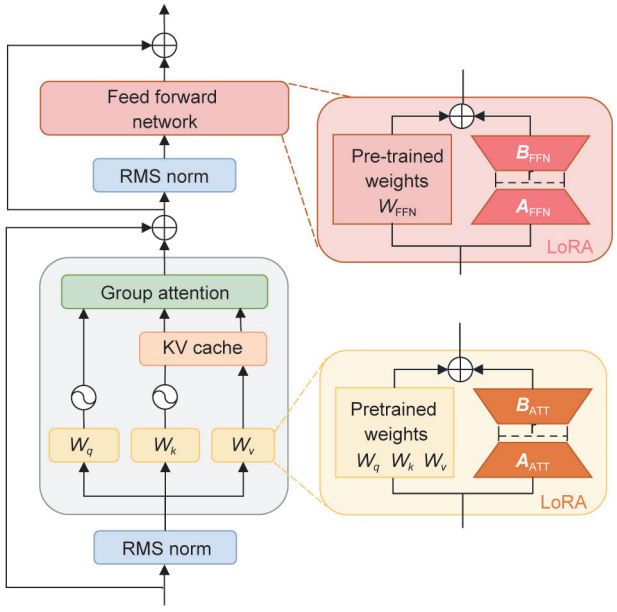


Fig. 6. One of the decoder blocks in our architecture design.

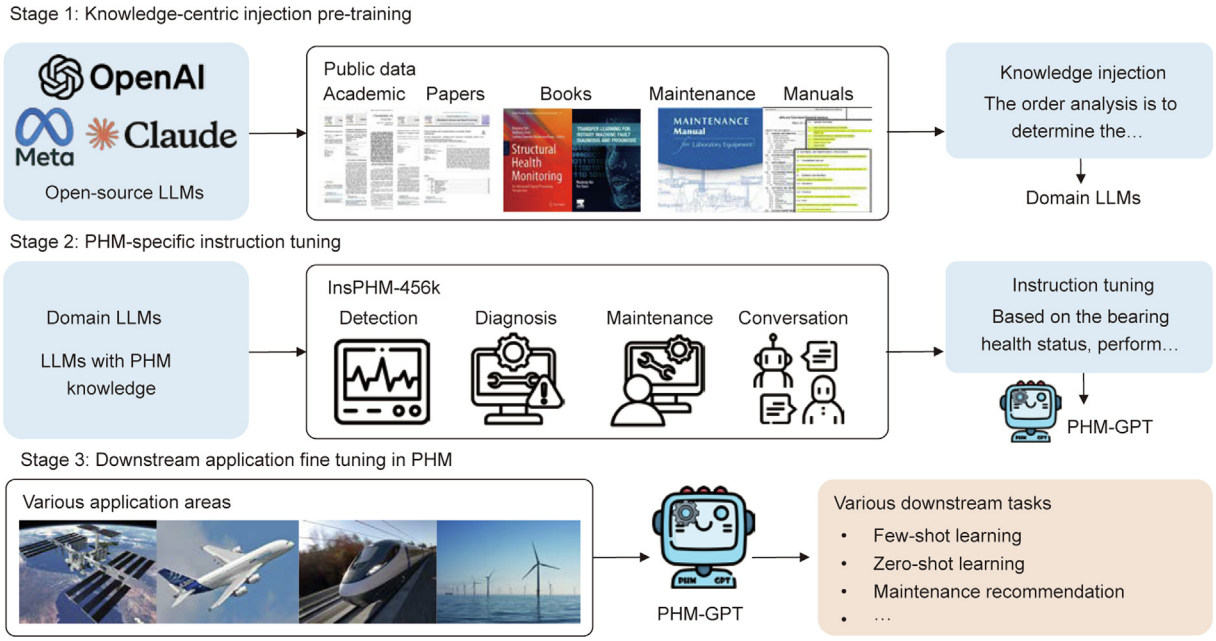


Fig. 7. Framework for generalizing the PHM–GPT from pretraining to downstream tasks.

PHM-related tasks. This stage uses the InsPHM-456k instruction dataset, which contains a large number of task-specific instructions. In contrast with the pretraining stage, which focuses on general PHM knowledge, this stage focuses on improving the ability of the model to perform different PHM tasks, including anomaly detection, fault diagnosis, maintenance decision-making, and PHM-related conversations. Through instruction tuning, the domain LLMs enriched with PHM knowledge are further adapted to the PHM–GPT, which is a specialized model that is capable of performing a wide range of PHM tasks.

Finally, in the fine-tuning stage, the PHM–GPT is applied across various industrial scenarios, including but not limited to aerospace, aviation, railways, and energy. Notably, instruction tuning focuses on enhancing the ability of the model to follow instructions, enabling it to respond flexibly to various PHM tasks. In contrast, fine-tuning focuses on the optimization of the performance achieved in downstream applications by adapting the model to specific domains, datasets, or operating conditions. When the PHM–GPT is adapted to new datasets, model pretraining and instruction tuning are not needed. Instead, only fine-tuning on the target dataset is necessary to achieve satisfactory performance, even under few-shot learning conditions. These results highlight the practical value of the PHM–GPT, which significantly enhances the efficiency of industrial predictive maintenance.

5. Simulation studies

In this section, a comprehensive evaluation of the PHM–GPT is conducted across a range of detection, diagnosis, and maintenance tasks. The simulation setup is first described, followed by an overview of the comparison models and evaluation metrics. The results of the simulation studies demonstrate the superiority of the PHM–GPT. To further assess the robustness of the PHM–GPT, a private dataset is employed to simulate realistic industrial scenarios involving previously unseen data. Moreover, the PHM–GPT is assessed under few-shot learning settings, highlighting its strong adaptability in situations with limited data. Finally, a case study is presented to demonstrate a practical deployment of the PHM–GPT, closely reflecting real industrial scenarios and completely differing from the existing DL-based PHM models.

5.1. Simulation study setup

All training and testing operations are conducted on a Linux system equipped with Nvidia A100 GPUs containing 80 GB of memory. The Python and PyTorch versions utilized are 3.11 and 2.3.1, respectively. The pretraining text is tokenized using the auto-tokenizer of Hugging Face, corresponding to the backbone model architecture. For both instruction tuning and fine-tuning, we adopt a LoRA adapter for the query, key, and value weights, which are essential for capturing the global dependencies in sequence modeling cases. A rank of 8 is selected for the LoRA modules, and a dropout rate of 0.1 is applied to mitigate overfitting. The cutoff length is 2048 tokens. Since our task involves causal language modeling, no input masking was applied. The batch size is set to a maximum of 40, and training was conducted for up to 50 epochs. During both the instruction tuning and the fine-tuning stages, we employ the AdamW optimizer with an initial learning rate of 10^{-4}. A cosine learning rate scheduler is employed to dynamically adjust the learning rate. To ensure fairness, the maximum number of new tokens, the temperature and the nucleus sampling parameter (top-p) are set to 512, 0.95, and 0.7, respectively, to encourage the generation of outputs that are both diverse and coherent.

A total of 114 236 instances are divided into training and testing sets at an 8:2 ratio. For each PHM task, these split instances are further used to construct instruction-based subdatasets for training and evaluation purposes. Moreover, the InsPHM-456k instruction dataset is inherently imbalanced because of the long-tailed distribution of its data sources, fault types, and rotational speeds. Rather than artificially rebalancing the data, we choose to preserve this natural distribution to better reflect the real-world imbalance that is commonly observed in industrial scenarios—thereby presenting a more realistic and meaningful challenge for the development and evaluation of PHM models.

To prevent data leakages, we construct a private dataset that is strictly isolated from the publicly available training dataset to further evaluate its generalization ability. As shown in Fig. 8, we conduct fault simulation experiments on a 1:1 testbed involving the main gearbox of a helicopter, which includes a drive motor, a load motor, a lubrication system, a main gearbox, a loading system, and a control system. Acceleration sensors are installed on the system, with a sampling frequency of 10 240 Hz. This dataset contains five bearing states, namely healthy, ball spalling, inner race spalling, outer race spalling, and cage cracking, which are used to evaluate the effectiveness of few-shot learning. We design simulations with 1, 5, and 10 shot configurations for each class to assess the ability of the model to generalize from limited labeled data.

5.2. Comparison models and evaluation metrics

To comprehensively evaluate our PHM–GPT, we conduct simulations involving a wide range of representative models. The time series models selected for comparison include LightTS [56], Transformer [57], Reformer [58], Autoformer [59], Cross-former [60], Informer [61], FEDformer [62], PatchTST [63], MICN [64], and TimesNet [65]. These models have achieved strong results in various time series tasks, including forecasting, detection, and classification. We also compare the PHM–GPT with several state-of-the-art LLMs, including Baichuan-2, Phi-1.5, Qwen-2.5, LLaMA-3, and DeepSeek-R1, to evaluate its instruction-following capabilities in the context of PHM tasks.

For the detection and diagnosis task, accuracy (ACC) is employed to assess the effectiveness of these models, offering a clear measure of their overall performance; this metric is calculated as follows:

$$\mathrm{ACC} = \frac{T_\mathrm{p} + T_\mathrm{n}}{T_\mathrm{p} + T_\mathrm{n} + F_\mathrm{p} + F_\mathrm{n}} \tag{4}$$

where T_p represents the number of positive samples that are correctly predicted as positive, T_n refers to the number of negative samples that are correctly predicted as negative, F_p refers to the number of negative samples that are incorrectly predicted as positive, and F_n refers to the number of positive samples that are incorrectly predicted as negative. However, while ACC offers a general assessment, it may not adequately reflect the performance of each model in specific scenarios, such as condition transfer tasks or cases with small sample sizes, underscoring the need for more evaluation metrics. To address this issue, additional metrics such as the recall (REC), precision (PREC), and F1 score (F1) are utilized, and are calculated as follows:

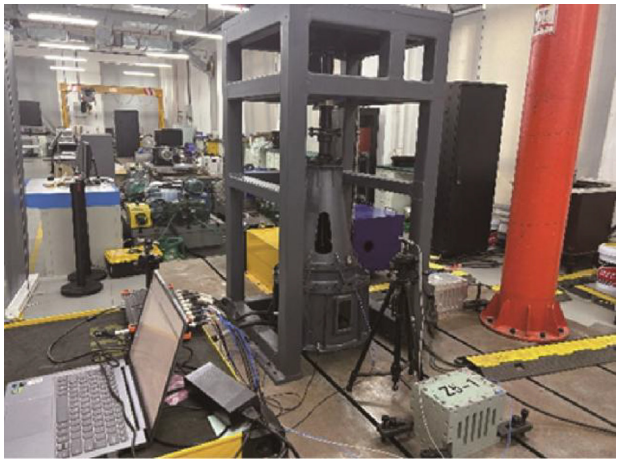


Fig. 8. Helicopter main gearbox fault simulation testbed.

$$\mathrm{REC} = \frac{T_\mathrm{p}}{T_\mathrm{p} + F_\mathrm{n}} \tag{5}$$

$$\mathrm{PREC} = \frac{T_\mathrm{p}}{T_\mathrm{p} + F_\mathrm{p}} \tag{6}$$

$$\mathrm{F1} = \frac{2 \times T_\mathrm{p}}{2 \times T_\mathrm{p} + F_\mathrm{p} + F_\mathrm{n}} \tag{7}$$

5.3. Anomaly detection case study

A comparison among the detection accuracies achieved across datasets is presented in Table 4. The PHM–GPT achieves outstanding results, with a total accuracy of 96.92%. It performs particularly well on the HIT, CWRU, and XJTU datasets, where its detection accuracies nearly reach 100%. Similarly, TimesNet and FEDformer achieve total accuracies of 93.37% and 92.49%, respectively. In contrast, the Transformer shows weaker performance on most datasets, particularly on DIRG and JNU, where its accuracies are less than 60%. Although it is a foundational model in sequence learning, it has several limitations, such as the inefficiency of full self-attention and the absence of a temporal inductive bias. Other advanced models overcome these issues through localized patch-based learning, frequency-domain representations, and multiscale architectural designs, resulting in improved accuracy and robustness.

A visual comparison among the performances achieved is shown in Fig. 9 through radar plots. Each plot illustrates the detection accuracy of a particular model across individual datasets. The plot of the PHM–GPT has a well-balanced shape, indicating uniformly high accuracy across all the datasets. In contrast, the other models display noticeable performance drops on specific datasets, likely because of the limited representation capacities and insufficient adaptability to diverse signal patterns. The superior performance of the LLM may be attributed to its effective integration of external knowledge, including parameters such as the rotational speed and sampling rate and meaningful signal features. These results highlight the potential of LLMs, such as the PHM–GPT, as reliable solutions for future anomaly detection challenges.

5.4. Fault diagnosis case study

To comprehensively evaluate the effectiveness of the PHM–GPT in diagnosis tasks and its applicability to real-world scenarios, traditional, downstream application fine-tuning, and ablation simulations are conducted.

5.4.1. Traditional simulation study

In the fault diagnosis task, a comparison among the diagnostic accuracies of various models across different datasets is presented in Table 5. The PHM–GPT achieves a total accuracy of 90.11%, significantly outperforming the other models. Specifically, the PHM–GPT achieves 100% accuracy on the XJTU dataset and maintains excellent performance across the other datasets, highlighting its robustness. Small models such as LightTS perform well on specific datasets such as XJTU and BJTU, but their total accuracies are lower than those of larger models such as Crossformer and FEDformer. Notably, models such as Autoformer and TimesNet exhibit significantly lower accuracies on certain datasets, which may be attributed to limited model capacity and biased classification boundaries. A key reason for this finding lies in the challenges posed by long-tailed data distributions in datasets, different fault types and various working conditions. Under such conditions, these models tend to overfit dominant patterns while neglecting rare but critical patterns. As a result, their learned representations fail to generalize

Table 4
Detection accuracy (%) comparison results obtained across different datasets in the anomaly detection task.

Model	CWRU	DIRG	HIT	HUB	IMS	JNU	PU	SDUB	SEUB	XJTU	UOC	HUG	SEUG	SDUG	BJTU	Total
LightTS	98.02	69.57	67.40	62.05	74.21	46.52	86.02	76.55	87.48	70.99	79.35	74.64	23.61	76.16	80.01	71.29
Transformer	90.76	46.16	67.75	78.66	71.74	53.22	75.57	79.12	79.00	89.06	88.06	65.79	79.64	81.41	86.02	77.32
Reformer	95.98	86.33	68.10	77.45	71.61	89.37	98.54	96.86	79.00	98.14	88.06	46.33	79.64	65.79	85.66	82.48
Autoformer	90.43	95.15	74.62	77.18	65.16	55.73	60.02	80.60	79.00	95.62	85.90	65.68	79.64	89.15	99.24	82.46
Crossformer	100.00	46.16	68.10	78.94	69.27	48.36	77.15	97.60	79.00	100.00	88.06	65.79	79.64	99.59	100.00	83.51
Informer	93.48	86.15	80.90	80.24	78.77	95.56	95.13	97.97	79.00	98.73	76.93	68.96	79.64	99.19	98.63	88.90
FEDformer	97.03	98.37	94.76	85.06	95.44	97.32	95.38	99.38	79.00	99.32	87.97	74.20	79.64	99.11	98.23	92.49
PatchTST	100.00	81.62	99.88	88.03	99.93	99.83	99.63	100.00	80.31	100.00	98.29	97.70	79.93	96.81	87.67	92.47
MICN	96.12	99.40	93.36	80.79	72.13	94.30	99.39	99.32	79.00	99.66	78.90	65.90	79.64	95.08	92.39	88.55
TimesNet	99.67	99.82	98.60	85.15	99.86	98.24	99.02	99.81	79.00	100.00	88.06	66.01	79.64	99.39	99.91	93.37
PHM–GPT	99.58	95.29	99.76	93.41	97.07	92.88	94.65	98.89	96.31	99.41	96.22	95.40	98.65	96.12	97.51	96.92

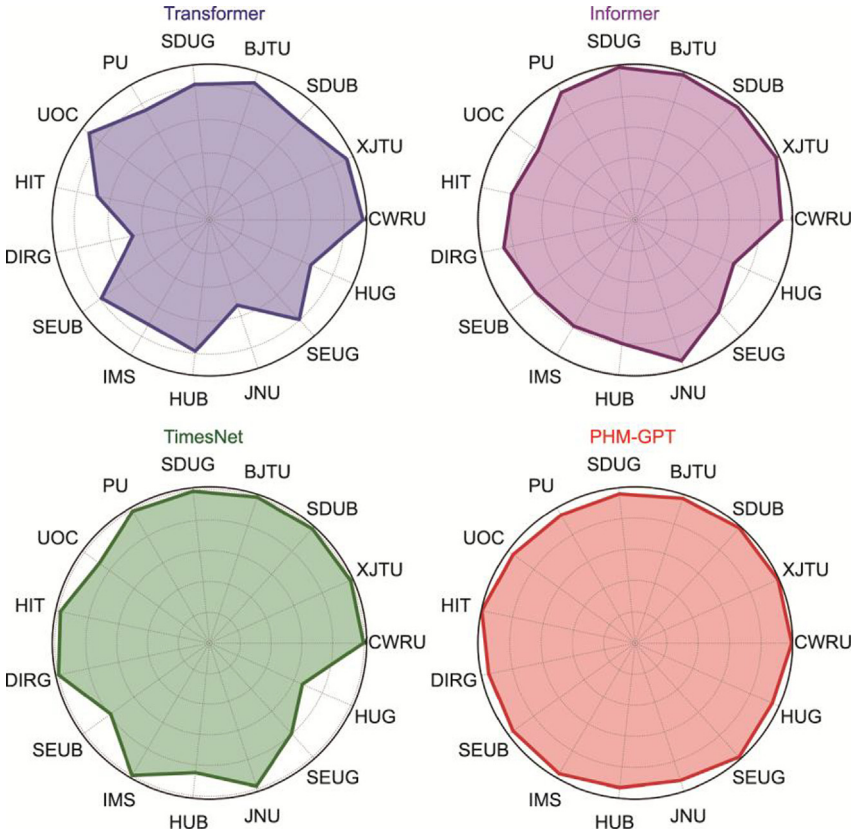


Fig. 9. Detection accuracy comparison among typical models in the anomaly detection task.

across the full spectrum of fault modes, limiting the adaptability of these models and leading to poor performance on underrepresented or complex datasets.

A comparison among the diagnostic accuracies of typical models is shown in Fig. 10. The transformer, Informer, and TimesNet suffer from obvious performance degradation on datasets such as SEUG. In contrast, the PHM–GPT achieves consistent performance across all the datasets. The presence of an extreme imbalance, where healthy samples dominate the training data and rare faults are severely underrepresented, often leads the conventional time series models to overfit the data. These results strongly support the potential of LLM-based models as robust solutions for handling the complexity and variability that are inherent in industrial fault diagnosis scenarios.

5.4.2. Downstream application fine-tuning simulation study

To better align with industrial applications, the PHM–GPT can be further adapted through fine-tuning on task-specific data. Utilizing the main gearbox of a helicopter as a representative example, we simulate a scenario where the PHM–GPT encounters previously unseen data to evaluate its generalization ability. As shown in Table 6, the PHM–GPT demonstrates remarkable few-shot learning performance. In the 1-shot setting, the PHM–GPT achieves an accuracy of 32.00%, which is competitive with those of the other models. In the 5-shot setting, the performance improves significantly to 64.00%, and in the 10-shot setting, the PHM–GPT reaches 70.44% accuracy. These results highlight the strong few-shot learning capabilities and superior generalization performance of the PHM–GPT. Its advantage can be largely attributed to the integration of unified instructions and effective features through the S2T pipeline, which enables the model to adapt quickly to new diagnostic tasks even with limited supervision.

Similar to other large models in few-shot learning settings, the proposed model may achieve satisfactory diagnosis performance in downstream applications even with limited training samples. One possible explanation is that the model has already effectively learned and internalized the underlying signal patterns by pre-training on sufficiently large and diverse datasets. Overall, the

Table 5
Diagnostic accuracy (%) comparison across different datasets in the fault diagnosis task.

Model	CWRU	DIRG	HIT	HUB	IMS	JNU	PU	SDUB	SEUB	XJTU	UOC	HUG	SEUG	SDUG	BJTU	Total
LightTS	61.00	82.22	63.32	21.98	82.55	61.17	61.23	72.92	20.99	96.29	16.60	57.15	20.35	89.43	95.61	63.49
Transformer	90.18	95.38	95.45	68.64	74.28	75.64	90.64	92.14	53.05	89.31	47.75	91.03	20.21	97.50	84.09	76.11
Reformer	98.76	99.91	99.76	66.69	98.69	99.49	100.00	99.93	20.89	100.00	96.67	76.17	23.75	99.87	95.58	82.58
Autoformer	96.37	99.74	91.73	59.74	97.85	98.74	99.87	98.71	19.28	99.37	82.31	71.47	20.01	99.11	96.81	80.33
Crossformer	100.00	99.93	99.89	70.68	99.54	98.07	100.00	99.93	46.54	100.00	98.20	90.92	26.67	99.91	99.72	86.64
Informer	95.05	99.91	98.60	61.59	98.50	98.99	100.00	99.81	40.73	99.83	96.05	80.10	36.49	99.63	99.44	85.64
FEDformer	99.34	99.82	99.06	65.21	98.76	99.58	99.51	99.63	36.14	99.91	97.21	93.87	37.26	99.91	99.72	86.42
MICN	99.67	99.40	99.83	85.25	98.69	99.58	99.63	99.14	55.02	99.57	98.47	99.23	32.66	95.80	79.82	85.23
PatchTST	92.82	90.94	79.39	76.99	93.88	83.84	81.28	87.90	90.66	97.98	89.76	95.08	84.19	94.23	86.05	88.64
TimesNet	100.00	99.91	99.06	70.87	99.21	98.66	98.78	99.63	59.91	100.00	99.73	95.08	20.16	99.71	98.77	87.19
PHM–GPT	95.54	96.92	99.18	82.46	98.50	78.49	98.29	88.95	97.07	100.00	88.33	99.98	75.91	82.86	89.29	90.11

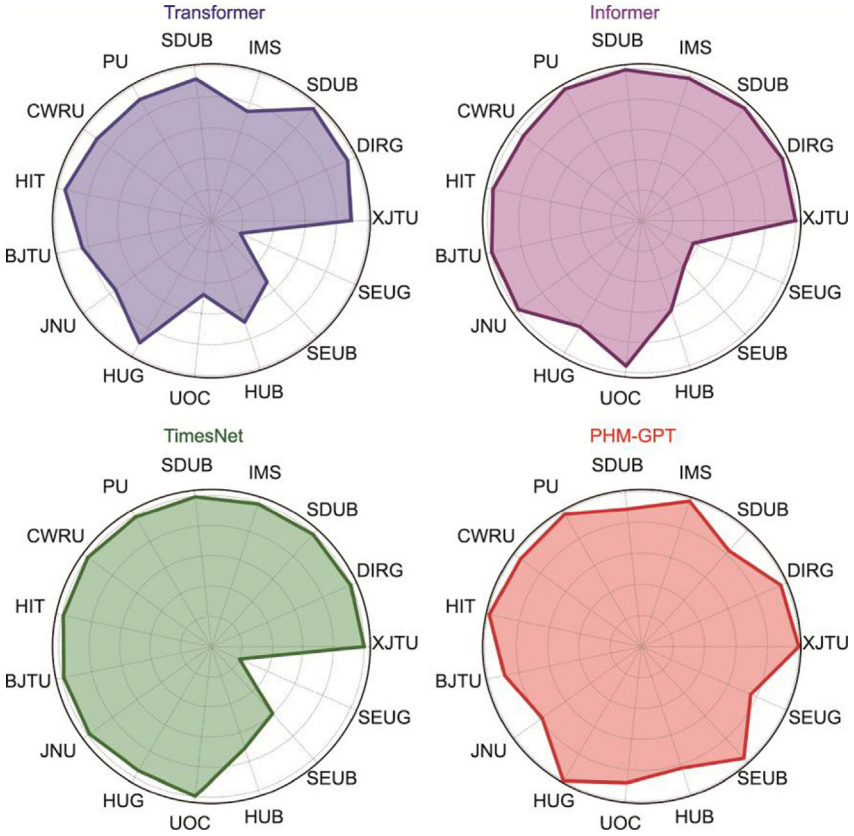


Fig. 10. Diagnostic accuracy comparison among typical models in the fault diagnosis task.

outstanding generalizability of the PHM–GPT makes it a highly effective tool for addressing few-shot learning tasks.

5.4.3. Ablation study

To further investigate the effectiveness of the S2T pipeline and instruction template design contained in the PHM–GPT, we conduct an ablation study by systematically removing the components of the instruction input. The instruction input integrates task-specific instructions, domain-related descriptions and statistical features derived from the S2T pipeline.

As shown in Table 7, removing the statistical features derived from the S2T pipeline (w/o S2T), leads to a dramatic decrease in performance, with the accuracy of the model plummeting to 21.98%, confirming the critical role of statistical features in guiding the model learning process. Excluding the domain-related descriptions (w/o Des) can result in a noticeable performance decline, while the removal of the task-specific instructions (w/o Ins) also weakens the model. These results indicate that instructions, descriptions, and statistical features are all indispensable. Their combination plays a crucial role in enhancing the understanding of PHM by the model, thereby validating the effectiveness of our instruction template design.

We also evaluate the influence of the number of features contained in the S2T pipeline by comparing settings with 6, 12, and 18, basic, and unique statistical features. As shown in Table 7 and Fig. 11, the performance of the model improves steadily with the addition of more features. The best results are achieved when all these statistical features are combined with the full instruction input, reaching 90.11% accuracy. These findings confirm the benefit of incorporating a larger number of statistical features, including domain-specific features. The integration of an advanced signal processing scheme in the S2T pipeline may further enhance the performance of the model by providing richer and more discriminative feature representations.

Table 6
Few-shot performance (%) comparison involving downstream application fault diagnosis tasks.

Model	1-shot				5-shot				10-shot			
	Accuracy	Recall	Precision	F1 score	Accuracy	Recall	Precision	F1 score	Accuracy	Recall	Precision	F1 score
LightTS	24.11	11.02	16.45	13.05	38.36	38.45	38.66	37.93	65.41	29.88	31.67	30.64
Reformer	20.96	11.11	10.81	10.65	44.03	44.69	44.74	43.19	52.41	52.39	52.10	50.57
Autoformer	25.79	24.90	30.55	23.67	48.43	47.99	46.89	39.22	55.56	55.39	56.03	47.95
Crossformer	22.64	15.43	14.26	14.39	50.52	50.95	53.46	49.87	61.22	60.79	60.99	60.15
MICN	29.14	27.98	26.64	26.93	55.14	54.99	56.09	54.73	64.36	65.28	70.74	63.93
PatchTST	29.98	29.44	30.39	28.65	54.09	52.97	60.18	52.31	66.04	64.79	67.34	64.83
TimesNet	32.91	31.58	41.49	29.40	61.84	61.15	62.25	61.40	68.13	67.39	68.89	67.38
PHM–GPT	32.00	34.85	31.09	32.40	64.00	62.09	61.75	61.35	70.44	69.89	70.89	69.51

To assess the influence of the LLM backbone, we replace the backbone of the PHM–GPT with several representative LLMs possessing different scales and architectures, including Phi-1.5, Qwen-2.5, LLaMA-3, and DeepSeek-R1. The results are summarized in Table 8. DeepSeek-R1 (14B) achieves the best performance, with an accuracy of 91.59%. In contrast, the smallest model Phi-1.5 (1.3B) yields noticeably lower results across all the metrics. These findings indicate that larger and advanced backbones significantly enhance the ability of the PHM–GPT to capture complex patterns and generalize across different datasets. Overall, this simulation highlights the importance of the selected LLM: a more powerful LLM facilitates a more effective representation learning procedure, a more accurate understanding of instructions and, ultimately, better performance in PHM scenarios.

Table 7
Ablation study results (%) obtained by removing components of the instruction input.

Input	Accuracy	Recall	Precision	F1 score
w/o S2T	21.98	13.80	16.53	12.48
w/o Des	55.43	57.41	55.39	48.51
w/o Ins	61.56	61.78	63.71	58.14
6	59.81	58.04	62.86	56.32
12	70.40	68.65	70.76	67.64
18	84.41	83.69	84.67	83.65
Basic	86.87	85.91	86.07	85.93
All	90.11	87.84	87.71	87.75

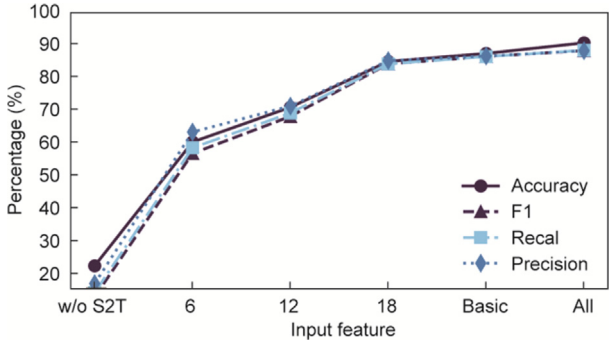


Fig. 11. Impact of the number of features contained in the S2T pipeline on the performance of the model.

Table 8
Language performance (%) comparison involving the maintenance task.

Backbone	Accuracy	Recall	Precision	F1 score
Phi-1.5	85.13	76.45	80.81	78.57
Qwen-2.5	88.27	84.31	83.27	83.79
LLaMA-3	90.11	87.84	87.71	87.75
DeepSeek-R1	91.59	89.97	89.66	89.74

5.5. Maintenance decision-making case study

In the maintenance decision-making case study, the model responses generated by several LLMs are investigated. Utilizing the inner fault maintenance task as an example, the responses derived from different LLMs are illustrated in Fig. 12. Baichuan-2 provides basic information about the bearings and emphasizes the importance of inspection. It focuses on checking several key indicators such as the temperature, pressure, and oil condition, but it does not offer specific guidance for the subsequent maintenance task.

Phi-3 focuses more on ensuring the safety of maintenance processes, including machinery and personnel safety, without providing specific maintenance recommendations. Although LLaMA-3 and Qwen-2.5 demonstrate strong logic, they focus primarily on investigating the root causes of bearing faults following the inspection process. In contrast, as shown in Fig. 12(e), the different responses acquired from the PHM–GPT offer more precise and executable maintenance guidance. Unlike general-purpose LLMs, the PHM–GPT leverages maintenance manuals, resulting in superior performance in specialized maintenance scenarios. However, this enhanced specialization comes at the cost of reduced generality, as compared with other LLMs, the PHM–GPT has fewer learnable parameters.

5.6. PHM process case study

A dialog example showing the end-to-end capabilities of the PHM–GPT in bearing anomaly detection, fault diagnosis and maintenance decision-making tasks is shown in Fig. 13. In this case study, monitoring indicators can be analyzed, the health status can be identified as “abnormal,” a fault can be diagnosed as an “inner fault,” and feasible maintenance recommendations can be provided. The PHM–GPT seamlessly integrates the anomaly detection, fault diagnosis, and maintenance decision-making tasks into a simplified dialog process.

In contrast, the traditional PHM methods often divide anomaly detection, fault diagnosis, and maintenance decision-making into isolated stages, resulting in low efficiency in equipment health management scenarios. For example, the diagnostic results yielded by DL models are difficult to directly translate into maintenance actions, necessitating references to maintenance manuals or the reliance on experience for repairs, which limits their economic value. Furthermore, the use of separate models for each PHM task increases the complexity of the system and slows maintenance decision-making processes. The PHM–GPT integrates various PHM tasks, reducing the levels of resource waste and downtime, thereby effectively advancing intelligent manufacturing.



Fig. 12. The responses derived from different LLMs in the bearing inner fault maintenance task. (a) Baichuan-2, (b) Phi-3, (c) LLaMA-3, (d) Qwen-2.5, and (e) PHM–GPT.

6. Discussions

As discussed in Section 5, the PHM–GPT performs well in PHM tasks such as anomaly detection, fault diagnosis, and maintenance decision-making, and outperforms models that are specifically trained on individual datasets. However, the transformative changes observed in other fields, which are caused by the emergence of LLMs, were not observed in the previous simulation studies. Unlike detection tasks that only input detection data and diagnosis tasks that only input diagnosis data, all data are input into the PHM–GPT. Currently, unpredictable emergent capabilities have begun to emerge, although there are still no means for fully controlling them. Therefore, this section discusses the progress achieved in areas, including attribution reasoning, threshold setting, and knowledge discovery, as well as the limitations. It is important to highlight that this aspect is intended for exploration, rather than the provision of simulation results.

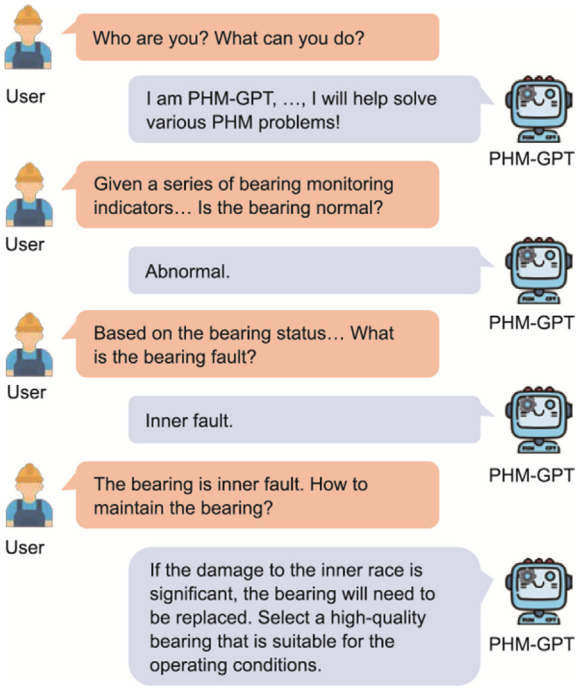


Fig. 13. The PHM process based on the PHM–GPT.

6.1. Attribution reasoning case study

Interpretability remains a significant challenge in the field of intelligent fault diagnosis. While DL models can make accurate predictions, their decision-making processes are often difficult for humans to intuitively comprehend. This "black box" nature can undermine users' trust in the obtained results, particularly in industrial applications. Currently, only a few specifically designed DL models offer partial interpretability. However, it seems easier for LLMs to achieve interpretability for PHM. In this case study, the PHM–GPT demonstrates some attribution reasoning ability, as shown in Table 9. During the inference process, the PHM–GPT not only demonstrates inference logic for faults on the basis of data but also attempts to uncover implicit patterns in mechanical system behaviors, such as degradation and wear.

To quantitatively assess attribution reasoning, we formulate root cause identification as a multilabel classification problem. Given a set of root causes generated by the LLM and expert-annotated ground truths, we compute precision as the proportion of correctly predicted causes among all the predicted causes. Since each case may involve multiple root causes, the evaluation is performed using micro-averaged precision. We compare the attribution reasoning outputs of the PHM–GPT with the expert annotations on 50 testing cases. The precision is 34.7%, and the diagnostic accuracy is 24.0%, indicating that the reasoning ability of the model is not yet fully mature. To address this limitation,

Table 9
Bearing fault question and answer case table.

Content type	Attribution reasoning case study	Threshold setting case study	Knowledge discovery case study
Question	You are an experienced prognostic and health management expert. What is the bearing fault?	You are an experienced prognostic and health management expert. What is the bearing fault?	You are an experienced prognostic and health management expert. What is the bearing fault?
Answer	A fascinating task! Given the comprehensive series of bearing status information and monitoring indicators, I'll perform a fault diagnosis task to determine the bearing's health. After analyzing the provided data, I notice some interesting patterns and deviations from the normal range. Here's a summary of my findings: 1. The mean, variance, standard deviation, and root amplitude are relatively small, indicating a stable signal. 2. The root mean square, peak-to-peak, and crest factor are moderate, suggesting a normal signal with some oscillations Considering these findings, I would diagnose the bearing fault as: **Inner fault** The combination of the moderate root mean square, peak-to-peak, and crest factor, along with the significant changes in the frequency domain (frequency mean, variance, and root mean square), suggests that the inner bearing components may be experiencing some degradation or wear. The large negative value of the seventh indicator and the very large value of the eighth indicator further support this diagnosis. Please note that this diagnosis is based on the provided data and may not be a definitive conclusion. A more comprehensive analysis, including additional data and expert judgment, would be necessary for a final diagnosis.	Based on the given bearing status information and monitoring indicators, I will perform a fault diagnosis task. First, I will analyze the indicators and identify any abnormalities. From the given data, I notice that the following indicators are outside the normal range: * Peak-to-peak value: 0.6289 (normal range: 0.1–0.3) * Root mean square value: 0.0934 (normal range: 0.01–0.1) * Crest factor: 3.4871 (normal range: 1–3) * Clearance factor: 5.151 (normal range: 1–3) * Impulse factor: 4.3785 (normal range: 1–3) These abnormal values suggest that the bearing may be experiencing a fault. Based on the analysis, I diagnose the bearing fault as: **Inner fault** The inner fault is characterized by abnormal values in the peak-to-peak, root mean square, crest factor, clearance factor, and impulse factor. These indicators suggest that the bearing's inner structure is damaged, leading to an imbalance in the bearing's vibration and noise patterns.	Based on the provided monitoring indicators, I will perform a fault diagnosis task. First, I will calculate the overall bearing health index (BI) using the following formula: **BI = (root amplitude/root mean square) × (1 − Kurtosis/3)** BI = (0.0571/0.0840) × (1 − 0.1858/3) BI = 0.6763 Next, I will analyze the skewness and kurtosis values: Skewness = 0.1618 (normal distribution: 0 < Skewness < 1.5) Kurtosis = 0.1858 (normal distribution: 0 < Kurtosis < 3) These values indicate that the distribution is slightly skewed and has a slightly higher kurtosis than normal Considering these findings, I would diagnose the bearing fault as: **Inner fault** The combination of the moderate root mean square, peak-to-peak, and crest factor, along with the significant changes in the frequency domain (frequency mean, variance, and root mean square), suggests that the inner bearing components may be experiencing some degradation or wear. The large negative value of the seventh indicator and the very large value of the eighth indicator further support this diagnosis. Please note that this diagnosis is based on the provided data and may not be a definitive conclusion. A more comprehensive analysis, including additional data and expert judgment, would be necessary for a final diagnosis.

constructing larger and more diverse expert-annotated datasets would be helpful for improving the alignment between the LLM outputs and expert judgments.

6.2. Threshold setting case study

Threshold setting in PHM is a critical step for distinguishing between normal and abnormal data; the threshold is typically determined in advance by experts through prior knowledge, statistical analyses, output distributions, and other factors. An appropriate threshold can improve the resulting detection accuracy but requires adjustments based on the data involved in specific applications. In this case study, the PHM–GPT demonstrates some threshold-setting ability, as shown in Table 9. This approach provides reasonable explanations that align closely with the classic fault diagnosis process, offering a threshold range for each monitoring feature. When a feature value exceeds its specified limit, it is considered abnormal. By comprehensively considering multiple indicators, the final diagnosis result can be determined.

To demonstrate the effectiveness of the thresholds given by the PHM–GPT, we sampled 10 instances of the same fault (inner race fault) from the testing dataset. The features of these instances, such as peak-to-peak value (P2P), root mean square value (RMS), crest factor (CF), clearance factor (CLF), and impulse factor (IF), are compared with the normal range inferred by the PHM–GPT. As shown in Fig. 14, the green shaded areas represent the normal ranges, while the yellow dots indicate the corresponding feature values of the instances. For several of the output features, the fault samples clearly differ from their normal threshold ranges, demonstrating the interpretability and effectiveness of the PHM–GPT in terms of threshold setting.

6.3. Knowledge discovery case study

The emergent capabilities of LLMs are particularly prominent in knowledge discovery tasks. If the previous two case studies aimed to make the traditional PHM process more intelligent and rational, knowledge discovery represents an effort to generate novel insights by proposing new knowledge for addressing the challenges encountered in PHM. The PHM–GPT demonstrates a strong knowledge discovery ability, as shown in Table 9. Without relying on any additional prior knowledge, the PHM–GPT leverages the mathematical capabilities of general-purpose LLMs to propose a bearing health indicator on the basis of the input monitoring indicators. With advancements in LLMs, it will become possible to uncover new knowledge, potentially transforming the existing PHM field.

To validate the bearing health index formula introduced by the PHM–GPT, we conducted a simulation on bearing 2–1 obtained from the XJTU-SY dataset. To improve the visualization of the trend, the BI values are negated. Although the BI lacks an empirical verification, our results demonstrate that it offers meaningful insights from a diagnostic perspective. As shown in Fig. 15, the raw signal remains relatively stable during the early stages of the bearing process, while the bearing health index (BI) remains close to zero. A significant increase in the BI is observed at the beginning of the fault, clearly distinguishing between the healthy state and the fault. Although the BI does not exhibit strong monotonicity over time, this is largely because the current focus of the PHM–GPT is limited to detection, diagnosis, and maintenance tasks. Nevertheless, the results are sufficient for confirming that the PHM–GPT can produce meaningful insights for PHM applications.

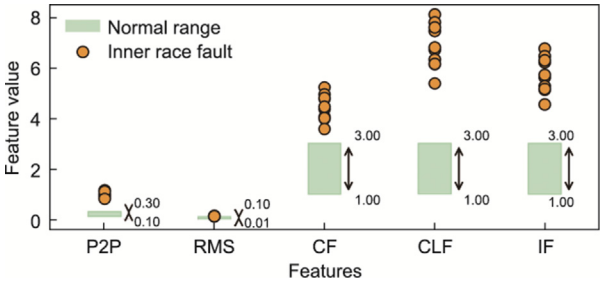


Fig. 14. Visualization of the thresholds produced in the threshold-setting case study.

6.4. Limitations

Although recent progress has been made in terms of leveraging the emergent capabilities of the PHM–GPT, several limitations remain.

First, LLMs are inherently prone to hallucination, generating outputs that may appear plausible but are factually incorrect. This issue is particularly critical in safety-sensitive applications in PHM. In the future, it will be essential to account for the inherent uncertainty of LLMs and to develop robust mechanisms that can mitigate their risks in critical decision-making scenarios.

Second, both the training and inference processes of LLMs typically require substantial computational resources. The traditional PHM models can usually be trained on a personal computer and perform inference tasks in milliseconds, making them well-suited for deployment in resource-constrained environments. In contrast, LLMs such as DeepSeek-R1 and LLaMA-3 require large amounts of GPU memory to work. As shown in Table 10, these models exhibit extensive computational requirements across multiple dimensions. Training a single model can take more than 40 h, and the process of obtaining an inference for one sample may exceed 20 s with DeepSeek. These computational demands present a

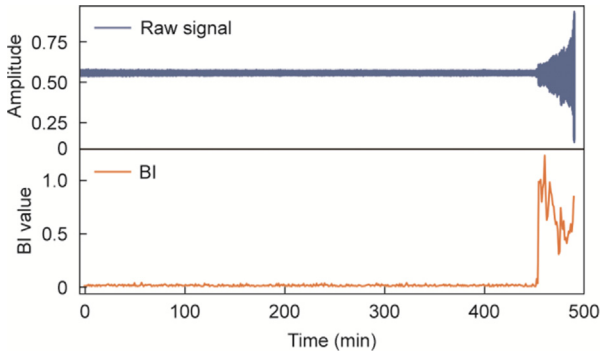


Fig. 15. Visualization of the BI values produced for XJTU-SY bearing 2–1.

Table 10
The computational resource requirements of different LLMs for diagnosis tasks.

LLM	Accuracy	Params (B)	Memory (GB)	Train (h)	Inference (s)
Phi-1.5	88.09	1.3	3	4	6.7
Qwen-2.5	88.27	7	16	21	12.9
LLaMA-3	90.11	8	17	25	13.4
DeepSeek-R1	91.59	14	34	49	21.3

Params denotes the number of model parameters, memory refers to the amount of GPU memory, train indicates the time required to train the associated model for 3 epochs on the diagnostic dataset, and inference represents the prediction time required for one sample with up to 512 output tokens.

significant challenge for deploying LLMs on edge devices or within latency-sensitive PHM applications, where both efficiency and responsiveness are critical. To address this limitation, future work should focus on lightweight designs, including techniques such as pruning, quantization, and knowledge distillation.

7. Conclusions and future work

In this paper, the PHM–GPT, a LLM that is designed specifically for the PHM field, is developed. A signal-to-text representation method and an instruction dataset construction method are introduced for the InstPHM-456k dataset. Additionally, a novel framework is proposed for adapting general-purpose LLMs to PHM-specific LLMs through knowledge injection-based pretraining, PHM-specific instruction tuning, and downstream application fine-tuning. By integrating various PHM tasks into an intuitive multidialog process, the PHM–GPT enables direct interactions with frontline employees. An extensive evaluation of the PHM–GPT reveals its state-of-the-art performance across various publicly available datasets, underscoring its robustness and versatility. Moreover, the PHM–GPT demonstrates emergent capabilities such as attribution reasoning, threshold setting, and knowledge discovery, holding the potential to transform the PHM field.

Although the PHM–GPT has achieved notable success, several areas warrant further exploration to broaden its application scope. First, integrating multimodal raw monitoring data into the instruction dataset would enable the PHM–GPT to address more complex PHM scenarios. Second, enhancing the interpretability of model outputs is critical for fostering the trust of frontline users in industrial settings. Third, extending the generalizability of the model to a wider range of tasks such as RUL prediction would significantly broaden its utility. Fourth, exploring lightweight deployment approaches would offer a promising pathway for optimizing the PHM–GPT for use in real-world industrial scenarios with limited computing resources. Finally, equipping the model with continual learning capabilities would allow it to adapt to changing operational conditions and new failure modes without the need for complete retraining. By addressing these areas, the PHM–GPT has the potential to become an important component of intelligent manufacturing, driving the digital transformation of industrial systems during the 4th industrial revolution.

CRediT authorship contribution statement

Jiaxin Ren: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Xue Liu:** Visualization, Methodology, Conceptualization. **Tianlei Wang:** Validation, Methodology, Conceptualization. **Zhibin Zhao:** Writing – review & editing, Supervision, Conceptualization. **Xuefeng Chen:** Writing – review & editing, Supervision, Conceptualization. **Weihua Li:** Writing – review & editing, Supervision, Conceptualization. **Ruqiang Yan:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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